Problem Set 2
ECO3121 - Fall 2022

October 28, 2022

Due 11.59pm, 10/27/2022

Question 1

1. Yes, there are other important determinants of crop yield which may also be correlated with fertilizer usage, including land quality, weather condition, farmers' personal characteristics, etc.

   Grading Rubric: 0.5 for the correct answer, and 0.5 for the explanation.

2. Suppose that the crop yield is positively affected by the farmers knowledge in agriculture, and that farmers with better knowledge tend to use more fertilizers. In this case, the use of fertilizer is likely to be positively correlated with farmers' knowledge leading to a positive value for the omitted variable bias so that $\hat{\beta}_1 > \beta_1$.

   Grading Rubric: 0.5 for the correct answer, and 0.5 for the explanation.

Question 2

1. gen Middle = 1 if sm==14|sm==36|sm==41|sm==42|sm==43

   replace Middle = 0 if Middle != 1

   sum Middle

   gen East = 1 if sm==33|sm==35|sm==44

   replace East = 0 if East != 1

   sum East

   gen West = 1 if sm==51

   replace West = 0 if West != 1

   sum West

gen Northwest = 1 if sm==22

replace Northwest = 0 if Northwest!= 1

sum Northwest

2. The regressor West is omitted to avoid perfect multicollinearity. If West is included, then the intercept can be written as a perfect linear function of the four regional regressors.

   Grading Rubric: 1 for the correct answer, 0.5 partial credit is on the right track.

3. The expected difference in earnings between Juanita and Jennifer is $-94.40 - (-12.89) = -81.55$.

   Grading Rubric: 0.5 for the correct answer.

4. **Testing against two-sided alternatives(Not Equals Zero)**

   - Choose a null hypothesis: $H_0 : \beta_{fertilizer} = 0$
   - Choose an alternative hypothesis: $H_1 : \beta_{fertilizer} \neq 0$
   - Choose a **significance level** for the test.
   - Calculate t-statistic $t_{\hat{\beta}_{fertilizer}} = \frac{\hat{\beta}_{fertilizer}}{se(\hat{\beta}_{fertilizer})}$
   - Choose a critical value, c>0, so that the **rejection rule** is

   $$t_{\hat{\beta}_{fertilizer}} > c$$

5. **Testing against two-sided alternatives (Coefficient equality)**

   - Choose a null hypothesis: $H_0 : \beta_{Middle} = \beta_{East}$
   - Choose an alternative hypothesis: $H_1 : \beta_{Middle} \neq \beta_{East}$
   - Choose a **significance level** for the test.
   - Run the regression
     $Yield = \beta_0 + \beta_1 Fertilizer + \beta_2 Area + \theta East + \beta_4(Middle + East) + \beta_5 Northeast$
     , where $\theta = \beta_3 - \beta_4$; and calculate the t-statistic $t_{\hat{\theta}} = \frac{\hat{\theta}}{se(\hat{\theta})}$
   - Choose a critical value, c>0, so that the **rejection rule** is

   $$t_{\hat{\theta}} > c$$

Question 3

1. 3.9%
2. $H_0 : \beta_2 = 0$ and $H_1 : \beta_2 > 0. t = \frac{0.00586}{0.001} = 5.86$. Critical value is 1.645. Because t-statistic is larger than critical value, so we reject the null hypothesis at the 5% significance level.
3. Lower. Because fewer independent variables in equation (3)
4. $\tilde{\beta}_1 = 0.039 + (0.00586)(3.534) = 0.0597$
5. Zero conditional mean of the error term. This is because IQ is in the error term and is correlated with education.

Question 4 is on the next page

6.11 (Requires calculus) Consider the regression model

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

for $i = 1, \ldots, n$. (Notice that there is no constant term in the regression.) Following analysis like that used in Appendix (4.2):

a. Specify the least squares function that is minimized by OLS.

b. Compute the partial derivatives of the objective function with respect to b_1 and b_2.

c. Suppose that $\sum_{i=1}^n X_{1i}X_{2i} = 0$. Show that $\hat{\beta}_1 = \sum_{i=1}^n X_{1i}Y_i / \sum_{i=1}^n X_{1i}^2$.

d. Suppose that $\sum_{i=1}^n X_{1i}X_{2i} \neq 0$. Derive an expression for $\hat{\beta}_1$ as a function of the data (Y_i, X_{1i}, X_{2i}), $i = 1, \ldots, n$.

e. Suppose that the model includes an intercept: $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$. Show that the least squares estimators satisfy $\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}_1 - \hat{\beta}_2 \overline{X}_2$.

f. As in (e), suppose that the model contains an intercept. Also suppose that $\sum_{i=1}^n (X_{1i} - \overline{X}_1)(X_{2i} - \overline{X}_2) = 0$. Show that $\hat{\beta}_1 = \sum_{i=1}^n (X_{1i} - \overline{X}_1)(Y_i - \overline{Y}) / \sum_{i=1}^n (X_{1i} - \overline{X}_1)^2$. How does this compare to the OLS estimator of β_1 from the regression that omits X_2?

a. $L = \sum (Y_i - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i})^2$

b. $\frac{\partial L}{\partial \hat{\beta}_1} = \sum -2X_{1i}(Y_i - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$

$\Rightarrow \hat{\beta}_1 = \frac{\sum X_{1i}Y_i - \hat{\beta}_2 \sum X_{1i}X_{2i}}{\sum X_{1i}^2}$ ①

$\frac{\partial L}{\partial \hat{\beta}_2} = \sum -2X_{2i}(Y_i - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$

$\Rightarrow \hat{\beta}_2 = \frac{\sum X_{2i}Y_i - \hat{\beta}_1 \sum X_{1i}X_{2i}}{\sum X_{2i}^2}$ ②

c. According to ①, $\hat{\beta}_1 = \frac{\sum X_{1i}Y_i}{\sum X_{1i}^2}$

d. Like ①

e. $L = \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i})^2$

$$\Rightarrow \frac{\partial L}{\partial \hat{\beta}_0} = \Sigma -(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$$

$$\Rightarrow \quad \bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2 = 0$$

$$\Rightarrow \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2$$

$$\text{f.} \quad \frac{\partial \Sigma (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i})^2}{\partial \hat{\beta}_1}$$

$$= \Sigma (-2 X_{1i})(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$$

$$\frac{\partial L}{\partial \hat{\beta}_2} = \Sigma (-2 X_{2i})(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0.$$

$$\Rightarrow \quad \overline{X_1 Y} - \hat{\beta}_0 \bar{X}_1 - \hat{\beta}_1 \overline{X_1^2} - \hat{\beta}_2 \overline{X_1 X_2} = 0$$

$$\overline{X_2 Y} - \hat{\beta}_0 \bar{X}_2 - \hat{\beta}_1 \overline{X_1 X_2} - \hat{\beta}_2 \overline{X_2^2} = 0.$$

$$\Rightarrow \begin{cases} \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} - \hat{\beta}_2 \bar{z} \\ \hat{\beta}_1 = \dfrac{Cov(x,y)\,Var(z) - Cov(z,y)\,Cov(x,z)}{Var(x)\,Var(z) - Cov(x,z)^2} \end{cases}$$

if $Cov(x,z) = 0$

$\Rightarrow \hat{\beta}_1 = \dfrac{Cov(x,y)}{Var(x)}$ equals $\hat{\beta}_1$ when x_2 is omitted.